

Database System Labs

- create db and import dataset

李旭东

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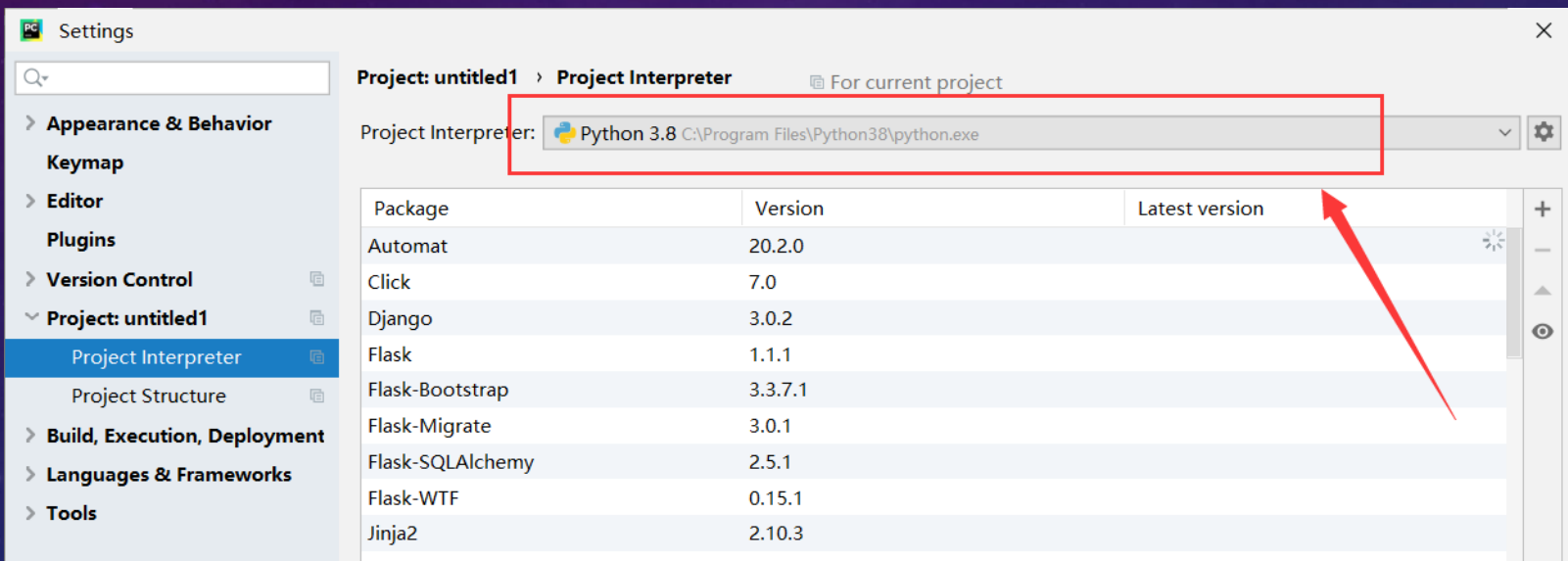
Nankai University

OBJECTIVES

- Python env config
- Create a new user and Grant DBA privilege
- Create a new DB and import some schemas
- SIMPLE QUERY STATEMENT
- quiz

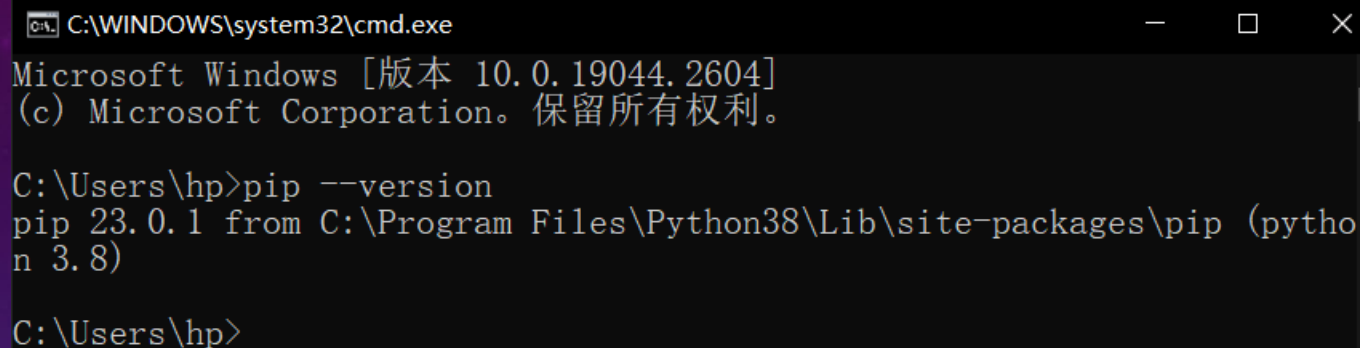
- Cmd中输入python能打出版版本号即安装成功,否则需要配置环境变量

```
C:\Users\hp>python
Python 3.8.0 (tags/v3.8.0:fa919fd, Oct 14 2019, 19:37:50) [MSC v.1916
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```



- Pycharm中File->Settings中选择好interpreter

- 确保拥有pip(cmd里输入pip)

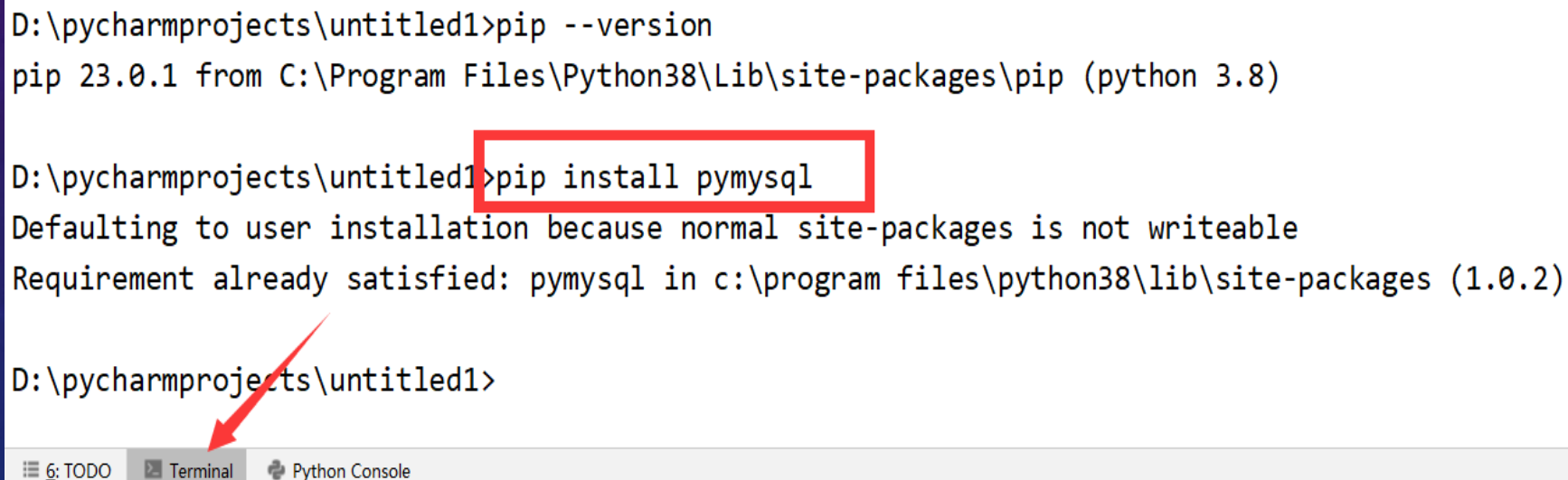


```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [版本 10.0.19044.2604]
(c) Microsoft Corporation。保留所有权利。

C:\Users\hp>pip --version
pip 23.0.1 from C:\Program Files\Python38\Lib\site-packages\pip (python 3.8)

C:\Users\hp>
```

- 在terminal中使用pip安装pymysql



```
D:\pycharmprojects\untitled1>pip --version
pip 23.0.1 from C:\Program Files\Python38\Lib\site-packages\pip (python 3.8)

D:\pycharmprojects\untitled1>pip install pymysql
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pymysql in c:\program files\python38\lib\site-packages (1.0.2)

D:\pycharmprojects\untitled1>
```

CONNECTION

```
import pymysql
conn=pymysql.connect(host='localhost',user='root',password='123',database='sys',charset='utf8')
cursor=conn.cursor()
sql = 'select version()'
cursor.execute(sql)
retdata = cursor.fetchall()
for row in retdata:
    print(row[0])
cursor.close()
conn.close()
```

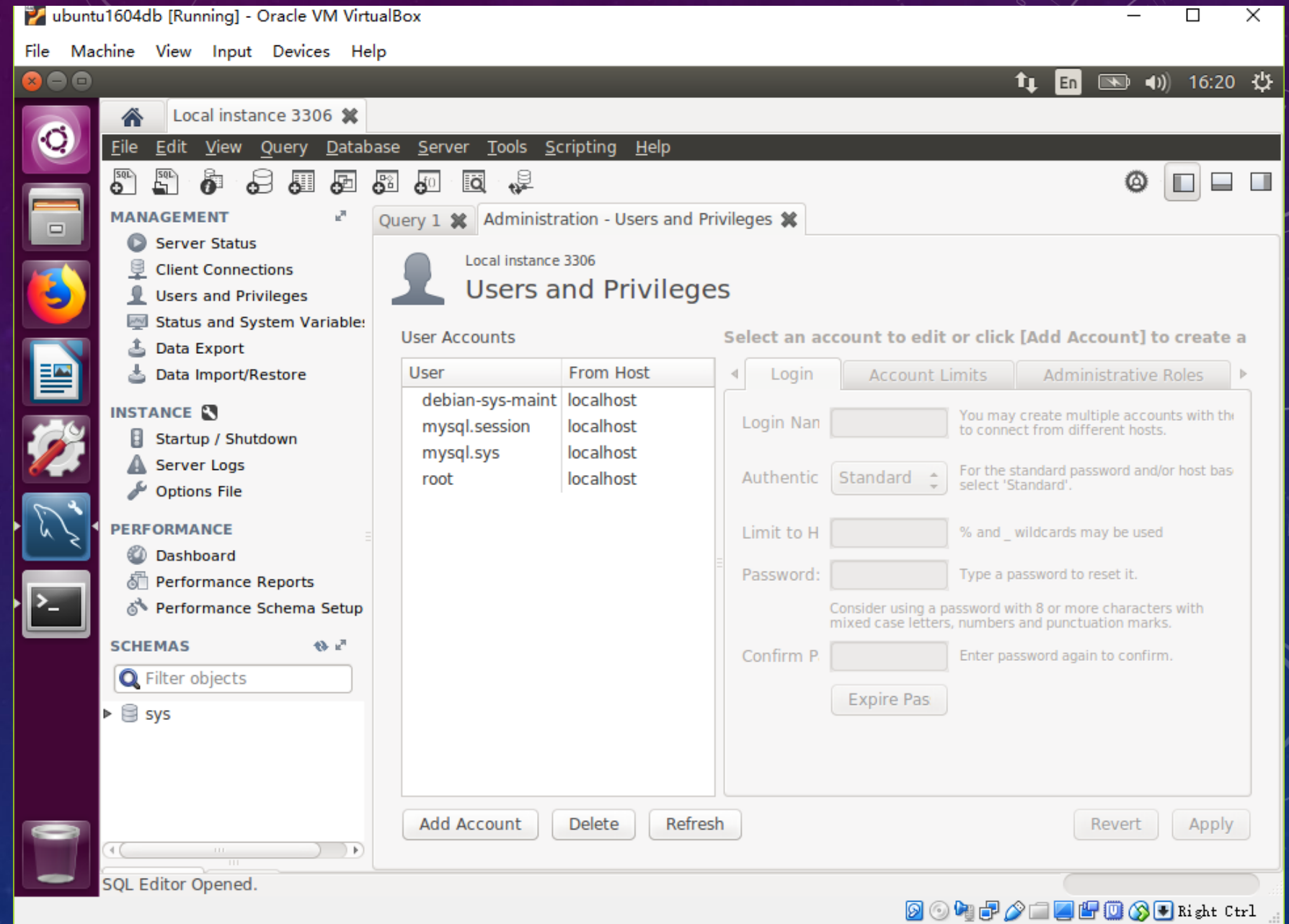
Database version : 8.0.19

Process finished with exit code 0

Create a new user and Grant DBA privilege

MYSQL: IMPORT DATA OF LABS

- Add DB User
 - myuser
 - mypwd



MYSQL: IMPORT DATA OF LABS

- Add DB User
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ubuntu1604db [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Local instance 3306

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variable
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

SCHEMAS

Filter objects

sys

Query 1 Administration - Users and Privileges

Local instance 3306

Users and Privileges

User Accounts

User	From Host
debian-sys-maint	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
newuser	%

Details for account newuser@%

Login Account Limits Administrative Roles

Login Name: myuser You may create multiple accounts with this name to connect from different hosts.

Authentication: Standard For the standard password and/or host based authentication select 'Standard'.

Limit to Host: % % and _ wildcards may be used

Password: Type a password to reset it.

Weak password.

Confirm Password: Enter password again to confirm.

Expire Password

Add Account Delete Refresh Revert Apply

SQL Editor Opened.

Right Ctrl

MYSQL: IMPORT DATA OF LABS

- Add DB User
 - myuser
 - mypwd

©LXD

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Users and Privileges

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mysql.session	localhost
mysql.sys	localhost
root	localhost
newuser	%

Details for account newuser@%

Login Account Limits Administrative Roles

Role

- ☒ DBA
- ☒ MaintenanceAdmin
- ☒ ProcessAdmin
- ☒ UserAdmin
- ☒ SecurityAdmin
- ☒ MonitorAdmin
- ☒ DBManager
- ☒ DBDesigner
- ☒ ReplicationAdmin
- ☒ BackupAdmin

Global Privileges

- ☒ ALTER
- ☒ ALTER ROUTINE
- ☒ CREATE
- ☒ CREATE ROUTINE
- ☒ CREATE TABLESPACE
- ☒ CREATE TEMPORARY TABLES
- ☒ CREATE USER
- ☒ CREATE VIEW
- ☒ DELETE
- ☒ DROP
- ☒ EVENT

Revoke All Privileges

Add Account Delete Refresh Revert Apply

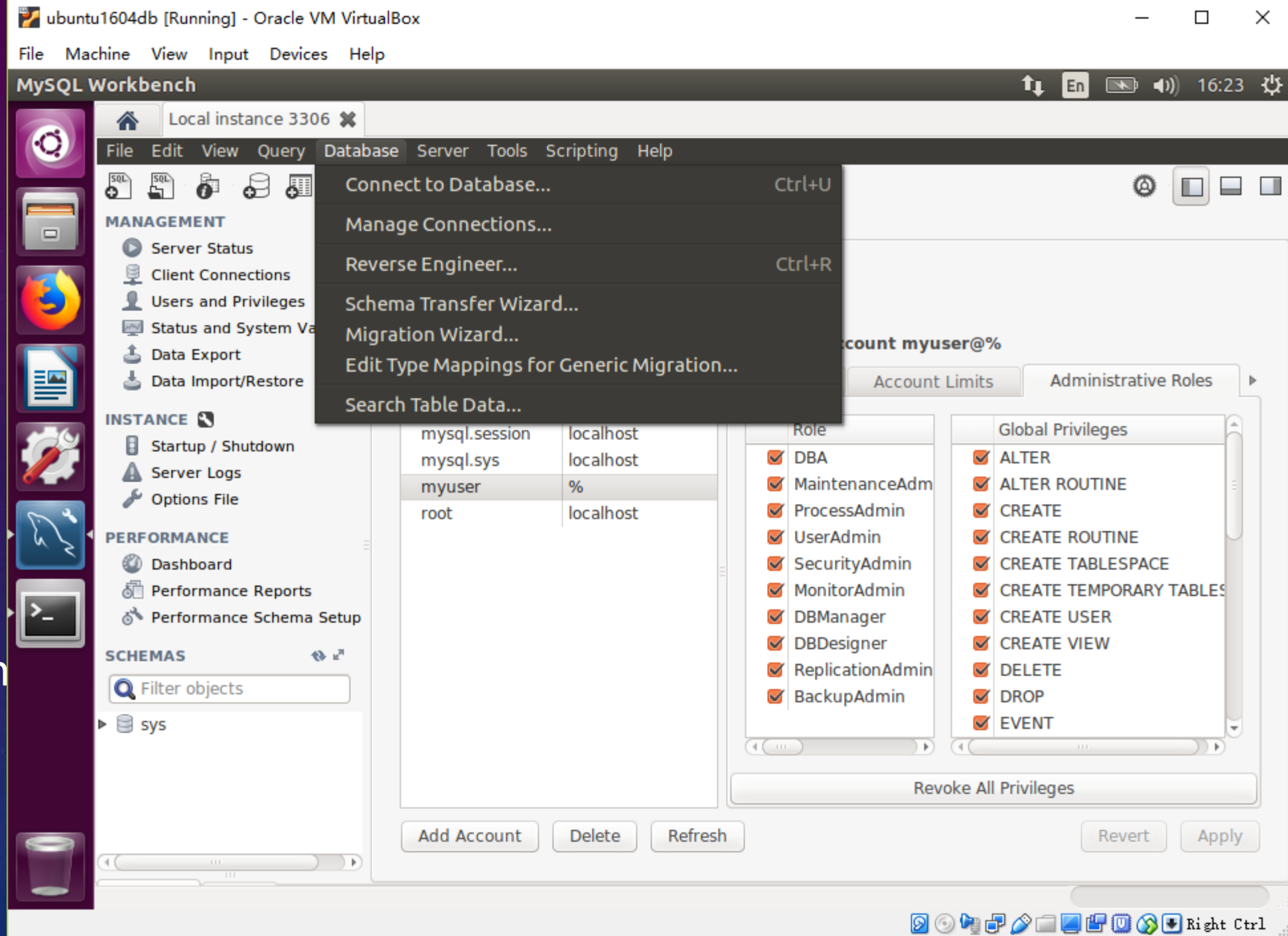
SQL Editor Opened.

Right Ctrl

MYSQL: IMPORT DATA OF LABS

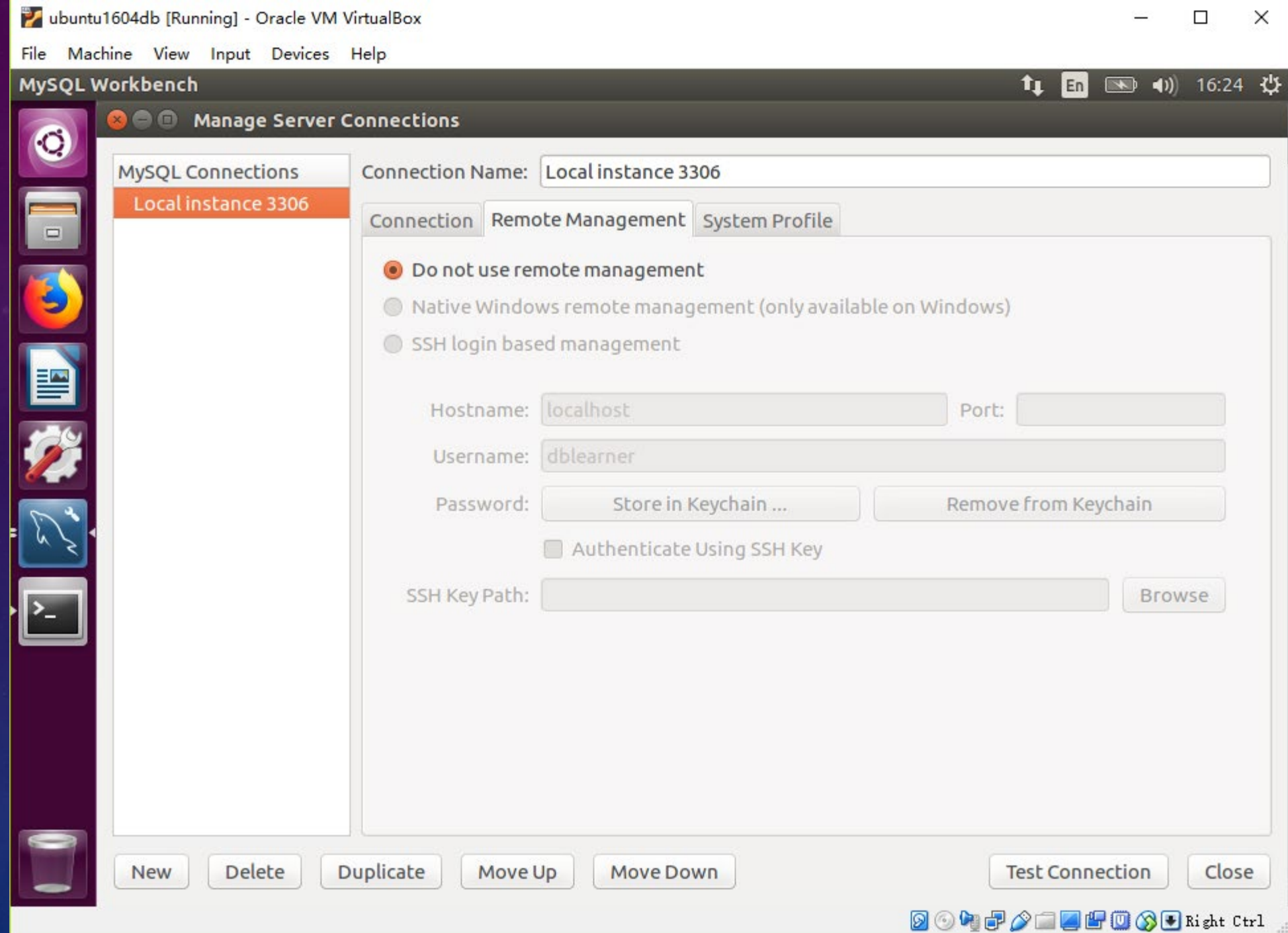
- New Connection

©LXD



MYSQL: IMPORT DATA OF LABS

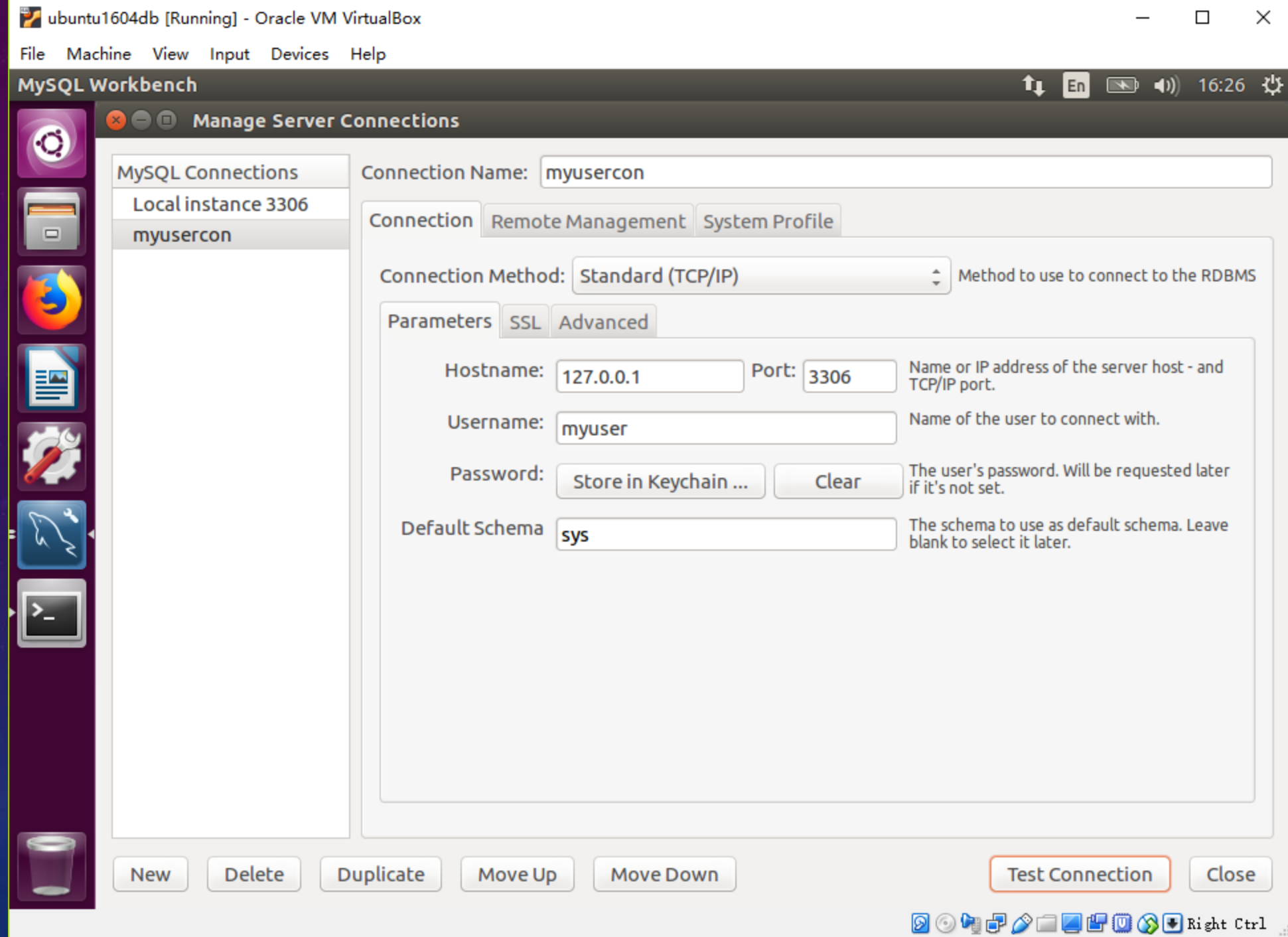
- New Connection



MYSQL: IMPORT DATA OF LABS

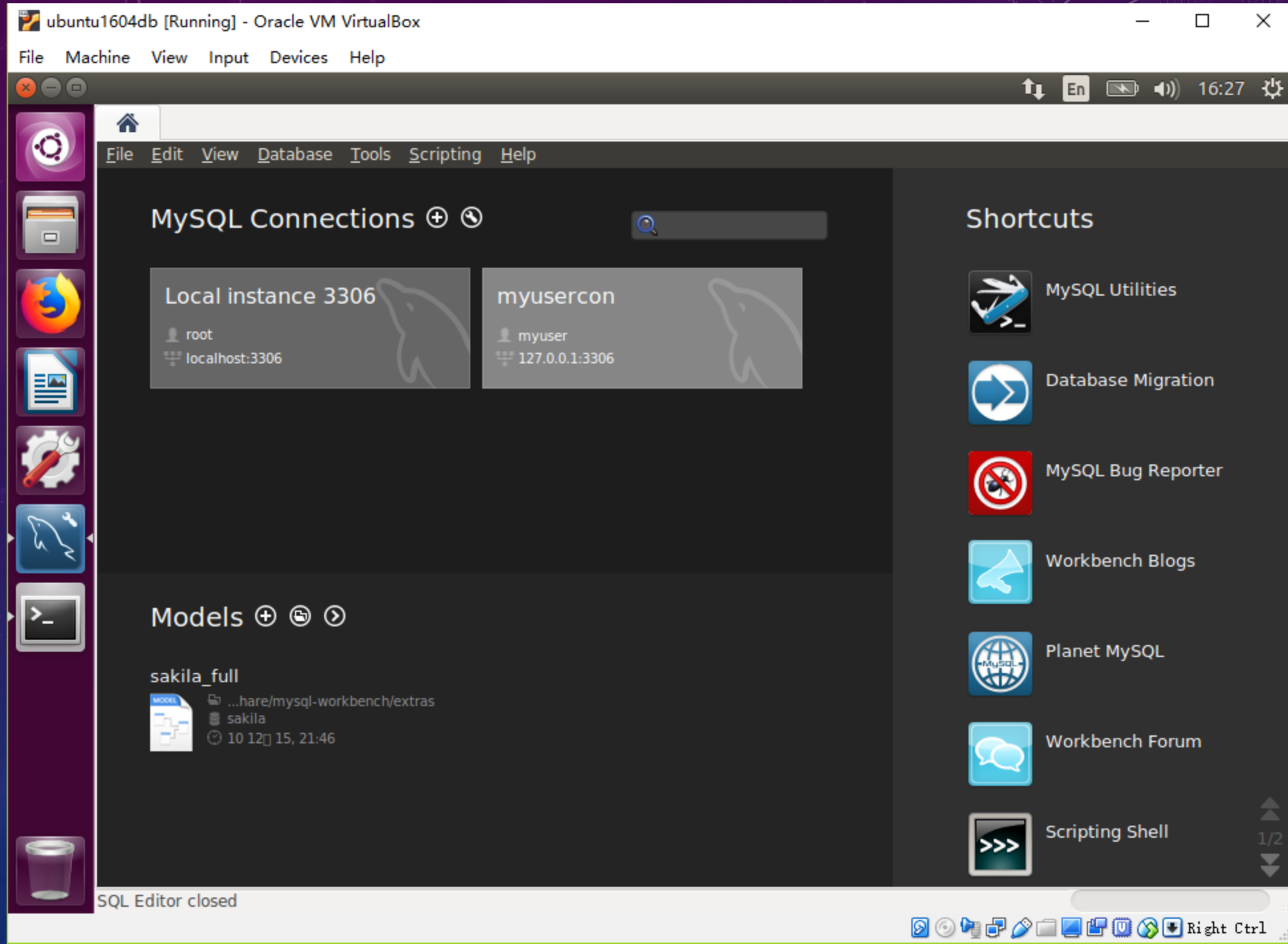
- New Connection

©LXD



MYSQL: IMPORT DATA OF LABS

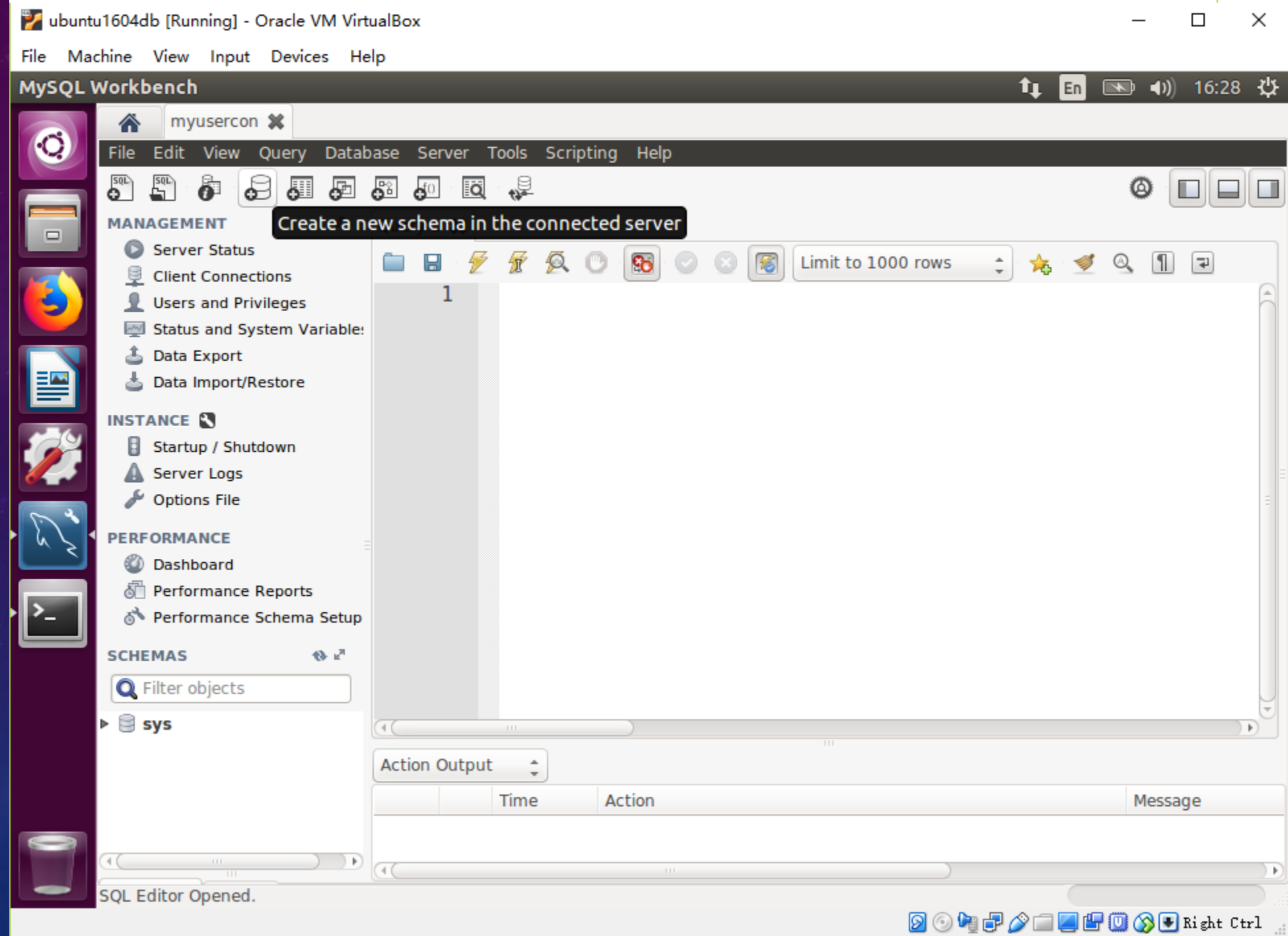
- New Connection



MYSQL: IMPORT DATA OF LABS

- New DB
 - dbsclab2018

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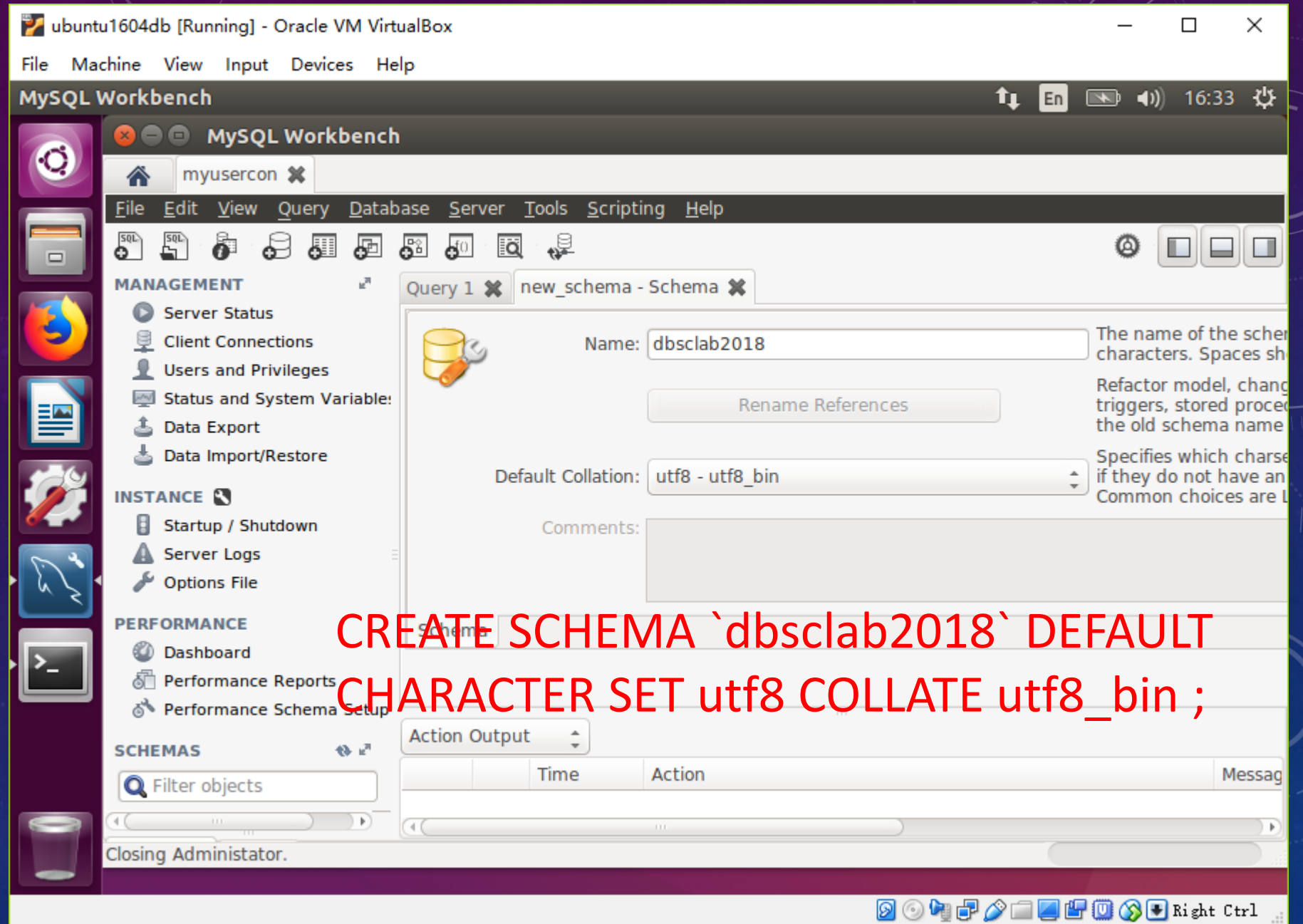




Create a new DB
and
import some schemas

MYSQL: IMPORT DATA OF LABS

- New DB
 - dbslab2018



CREATE SCHEMA `dbslab2018` DEFAULT
CHARACTER SET utf8 COLLATE utf8_bin ;

MySQL Workbench interface showing the 'Users and Privileges' configuration for the 'myusercon' database.

MANAGEMENT

- Server Status
- Client Connections
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- Data Export
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INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

SCHEMAS

Filter objects

- dbslab2018
- sys
 - Tables
 - Views
 - Stored Proc
 - Functions

Users and Privileges

User Accounts

User	From Host
debian-sys-maint	localhost
mysql.session	localhost
mysql.sys	localhost
myuser	%
root	localhost

Details for account myuser@%

Login | Account Limits | Administrative Roles | Schema Privileges

Login Name: myuser
You may create multiple accounts with the same name to connect from different hosts.

Authentication Type: Standard
For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching: %
% and _ wildcards may be used

Password:
Type a password to reset it.

Consider using a password with 8 or more characters with mixed case letters, numbers and punctuation marks.

Confirm Password:
Enter password again to confirm.

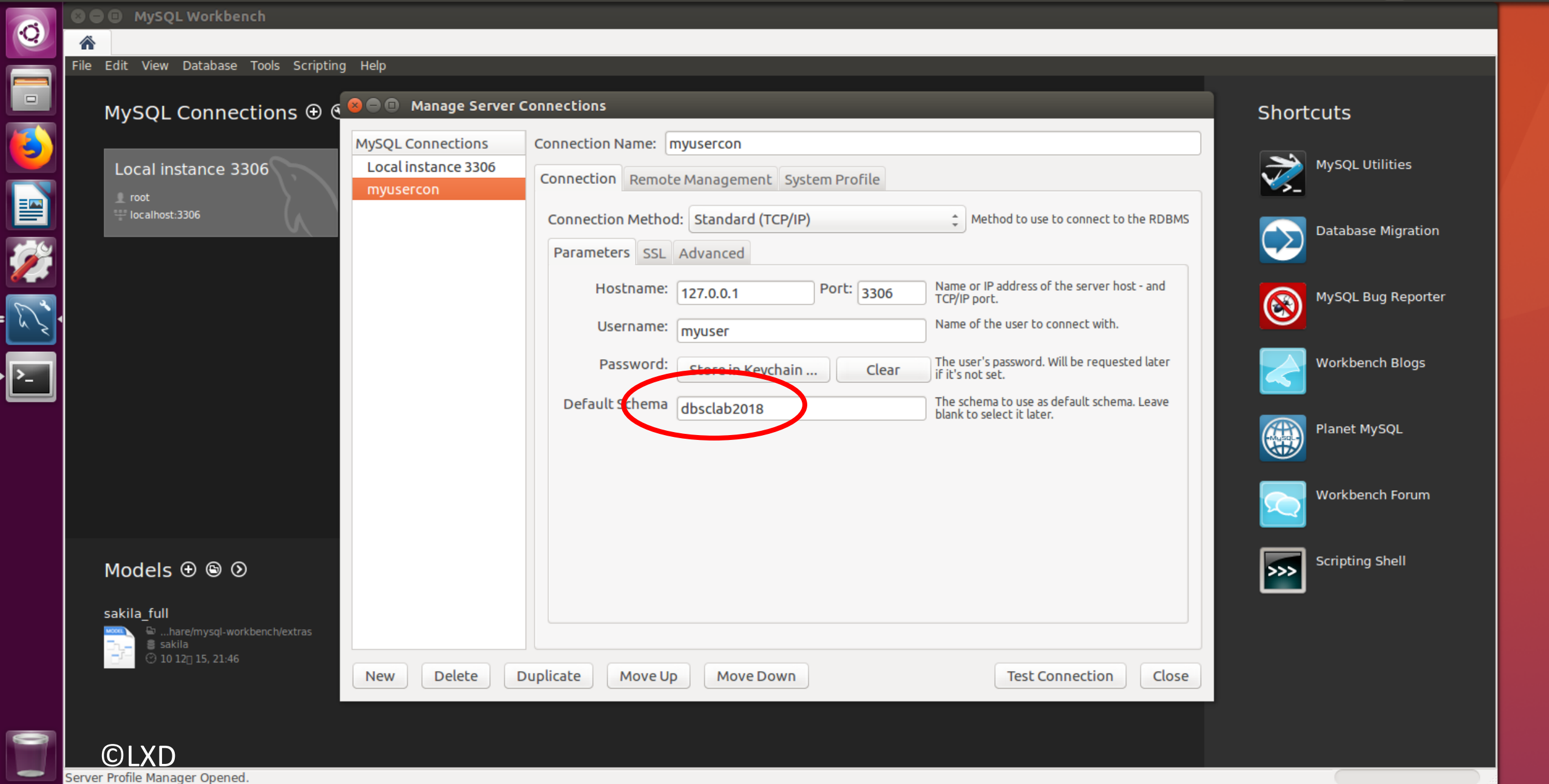
Expire Password

Delete Refresh

Revert Apply

Context Menu:

- Set as Default Schema
- Filter to This Schema
- Schema Inspector
- Table Data Import Wizard
- Copy to Clipboard
- Send to SQL Editor
- Create Schema...
- Alter Schema...
- Drop Schema...
- Search Table Data...
- Refresh All



The screenshot shows the MySQL Workbench interface with the 'Manage Server Connections' dialog box open. The dialog has three tabs: 'Connection', 'Remote Management', and 'System Profile'. The 'Connection' tab is active, showing the 'Parameters' sub-tab. The 'Connection Name' is 'myusercon'. The 'Connection Method' is 'Standard (TCP/IP)'. The 'Hostname' is '127.0.0.1' and the 'Port' is '3306'. The 'Username' is 'myuser'. The 'Password' field has a 'Store in Keychain ...' button and a 'Clear' button. The 'Default Schema' is 'dbslab2018', which is circled in red. The 'Test Connection' button is visible at the bottom right of the dialog.

MySQL Connections

Local instance 3306

root
localhost:3306

Models + - >

sakila_full

...hared/mysql-workbench/extras
sakila
10 12 15, 21:46

©LXD

Server Profile Manager Opened.

Shortcuts

- MySQL Utilities
- Database Migration
- MySQL Bug Reporter
- Workbench Blogs
- Planet MySQL
- Workbench Forum
- Scripting Shell

Manage Server Connections

Connection Name: myusercon

Connection Remote Management System Profile

Connection Method: Standard (TCP/IP) Method to use to connect to the RDBMS

Parameters SSL Advanced

Hostname: 127.0.0.1 Port: 3306 Name or IP address of the server host - and TCP/IP port.

Username: myuser Name of the user to connect with.

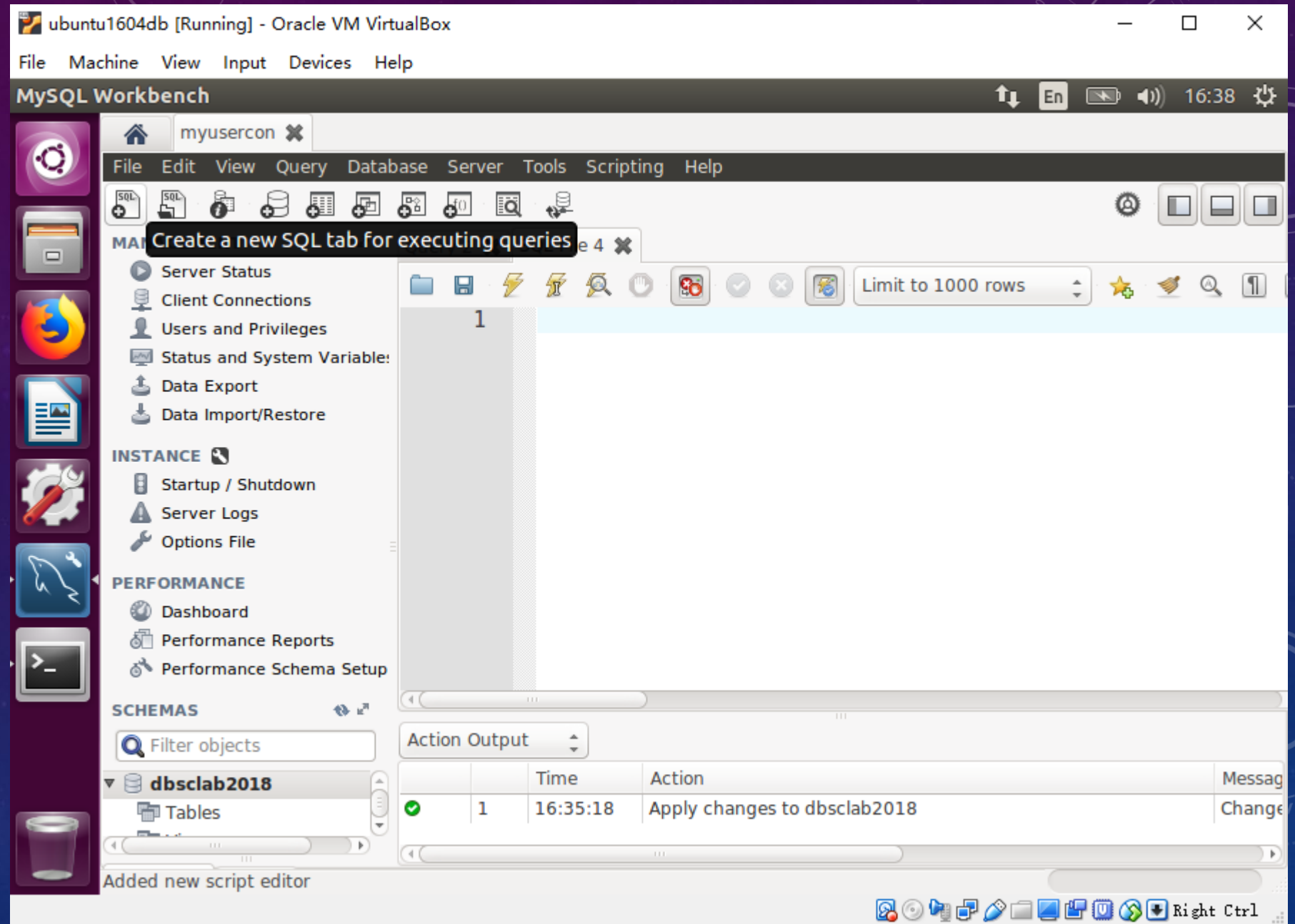
Password: Store in Keychain ... Clear The user's password. Will be requested later if it's not set.

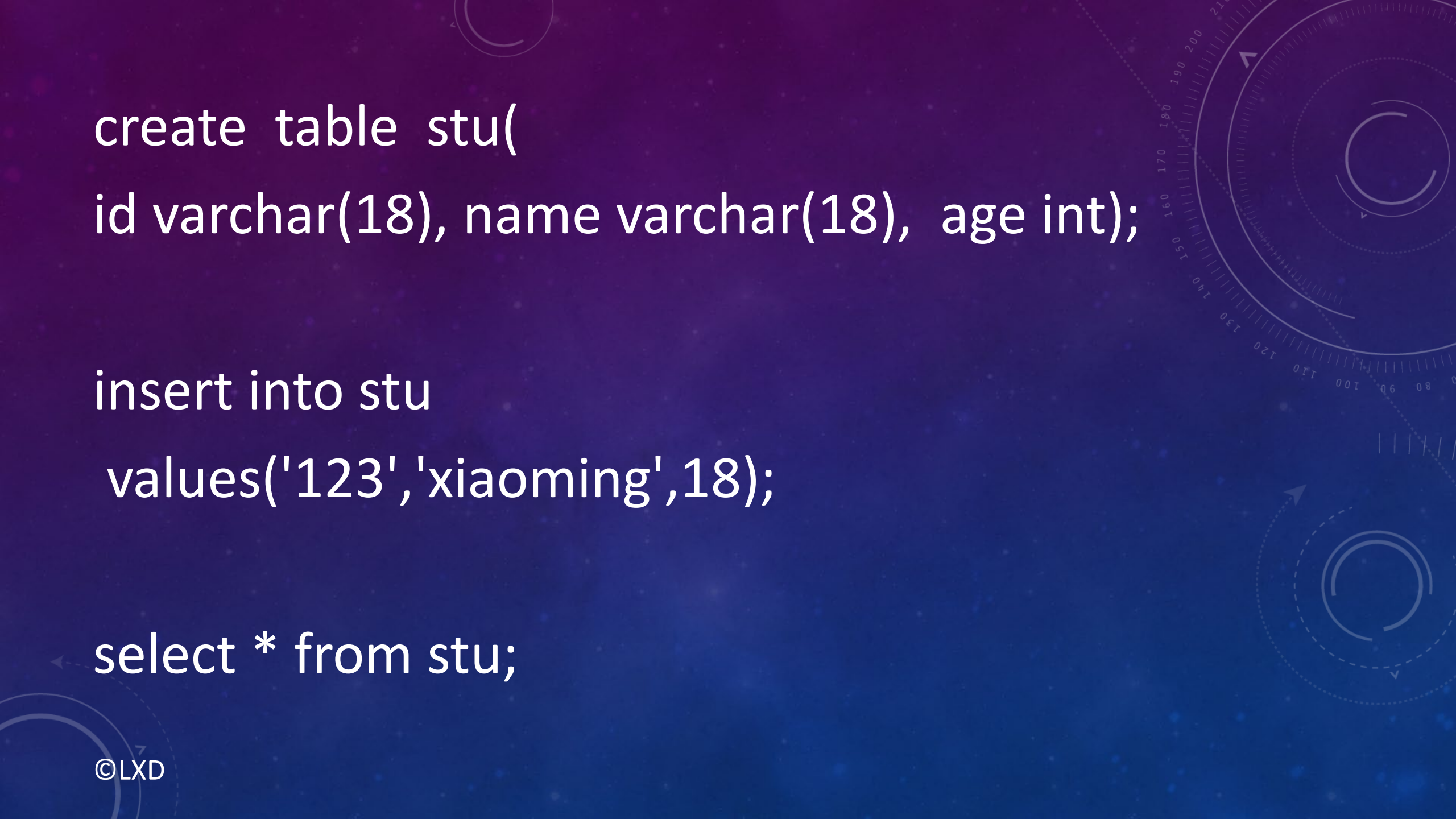
Default Schema dbslab2018 The schema to use as default schema. Leave blank to select it later.

New Delete Duplicate Move Up Move Down Test Connection Close

MYSQL: IMPORT DATA OF LABS

- New DB
 - dbslab2018





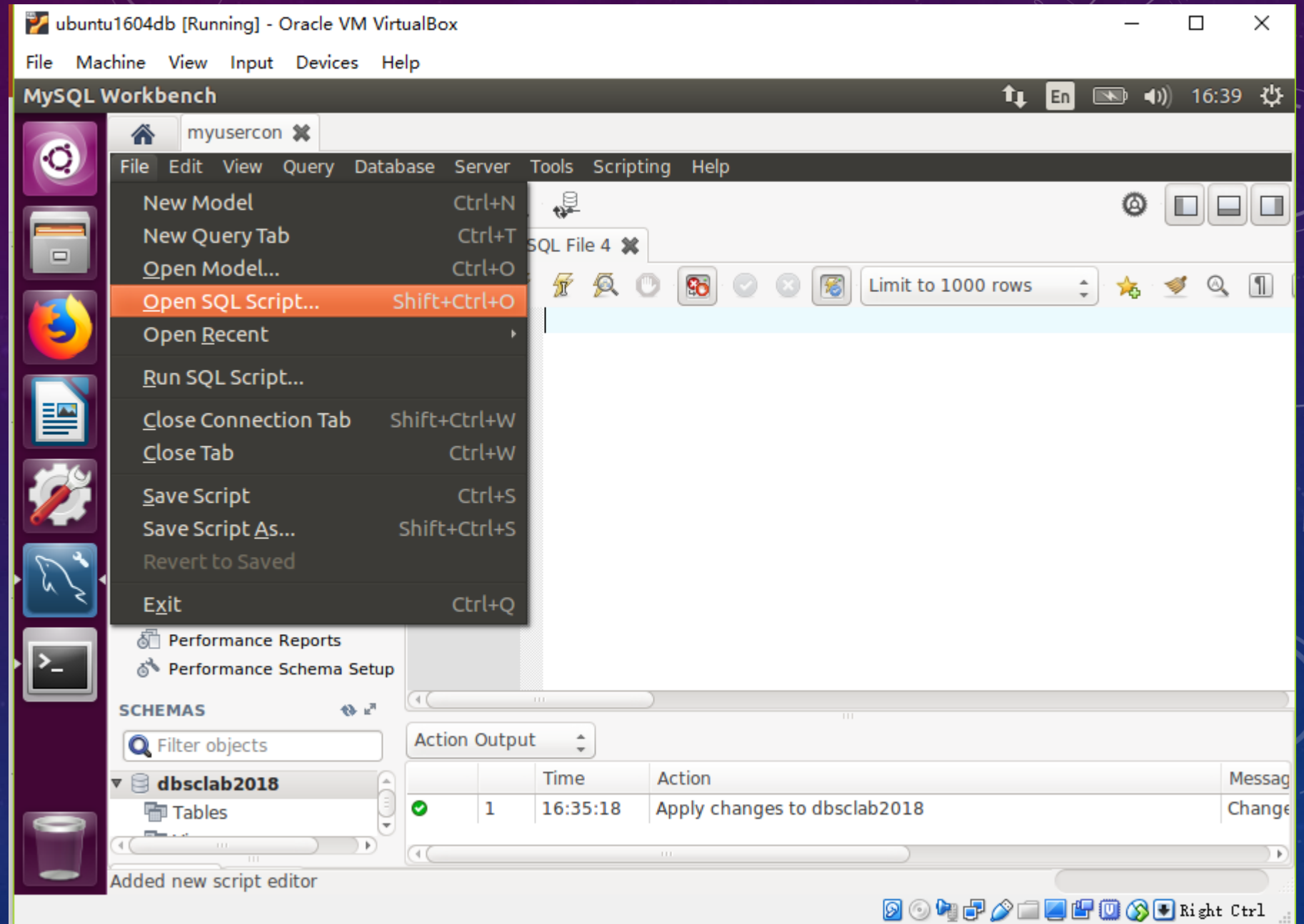
```
create table stu(  
id varchar(18), name varchar(18), age int);
```

```
insert into stu  
values('123','xiaoming',18);
```

```
select * from stu;
```

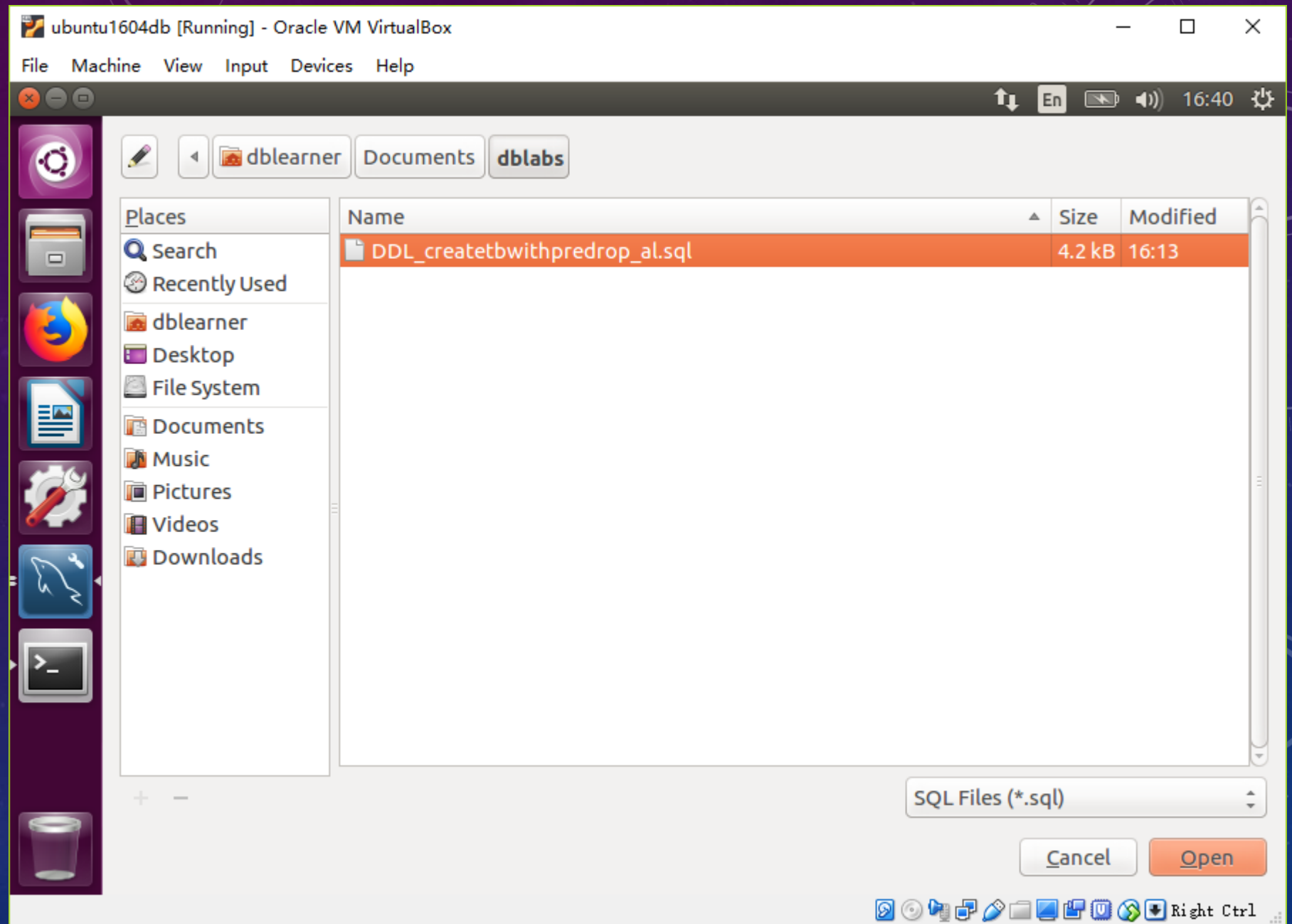
MYSQL: IMPORT DATA OF LABS

- New DB
 - dbslab2018



MYSQL: IMPORT DATA OF LABS

- New DB
 - dbsclab2018



ubuntu1604db [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

myusercon

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

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PERFORMANCE

- Dashboard
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- Performance Schema Setup

SCHEMAS

Filter objects

db sclab2018

- Tables
 - advisor
 - classroom
 - course
 - department
 - instructor
 - prereq
 - section
 - student
 - takes
 - teaches

Query 2 SQL File 4 DDL_createtbwithpredrop_al

Limit to 1000 rows

```
31 credits numeric(2,0) check (credits > 0),
32 primary key (course_id)
33 );
34
35 • alter table `course` add constraint `course_fk1`
36   foreign key (`dept_name`) references `department` (`dept_name`)
37   on delete set null;
38
39 • create table instructor
40   (ID varchar(5),
41    name varchar(20) not null,
42    dept_name varchar(20),
43    salary numeric(8,2) check (salary > 29000),
44    primary key (ID)
45   );
46
47 • alter table `instructor` add constraint `instructor_fk1`
48   foreign key (`dept_name`) references `department` (`dept_name`)
49   on delete set null;
50
51 create table section
```

Action Output

	Time	Action	Message	Duration / Fetch	
✓	29	16:40:42	alter table `takes` add constraint `takes_fk2` foreign key (ID...	Records: 0 Duplicates: 0 Warnings: 0 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.036 sec
✓	30	16:40:42	create table advisor (s_ID varchar(5), i_ID...	0 row(s) affected	0.015 sec
✓	31	16:40:42	create table time_slot (time_slot_id varchar(4), ...	0 row(s) affected	0.011 sec
✓	32	16:40:42	create table prereq (course_id varchar(8), pre...	0 row(s) affected	0.013 sec

Query Completed

Right Ctrl

MYSQL: IMPORT DATA OF LABS

- New DB
 - dbsclab2018

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ubuntu1604db [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

MySQL Workbench

myusercon X

File Edit View Query Database Server Tools Scripting Help

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PERFORMANCE

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SCHEMAS

Filter objects

- departments
- instructor
 - Columns
 - ID

Query 2 X DDL_createtbwithpredrop_al X

Limit to 1000 rows

```
1 • select ID,name,dept_name from instructor;
```

Result Grid

#	ID	name	dept_name
*	NULL	NULL	NULL

instructor 3 X

Action Output

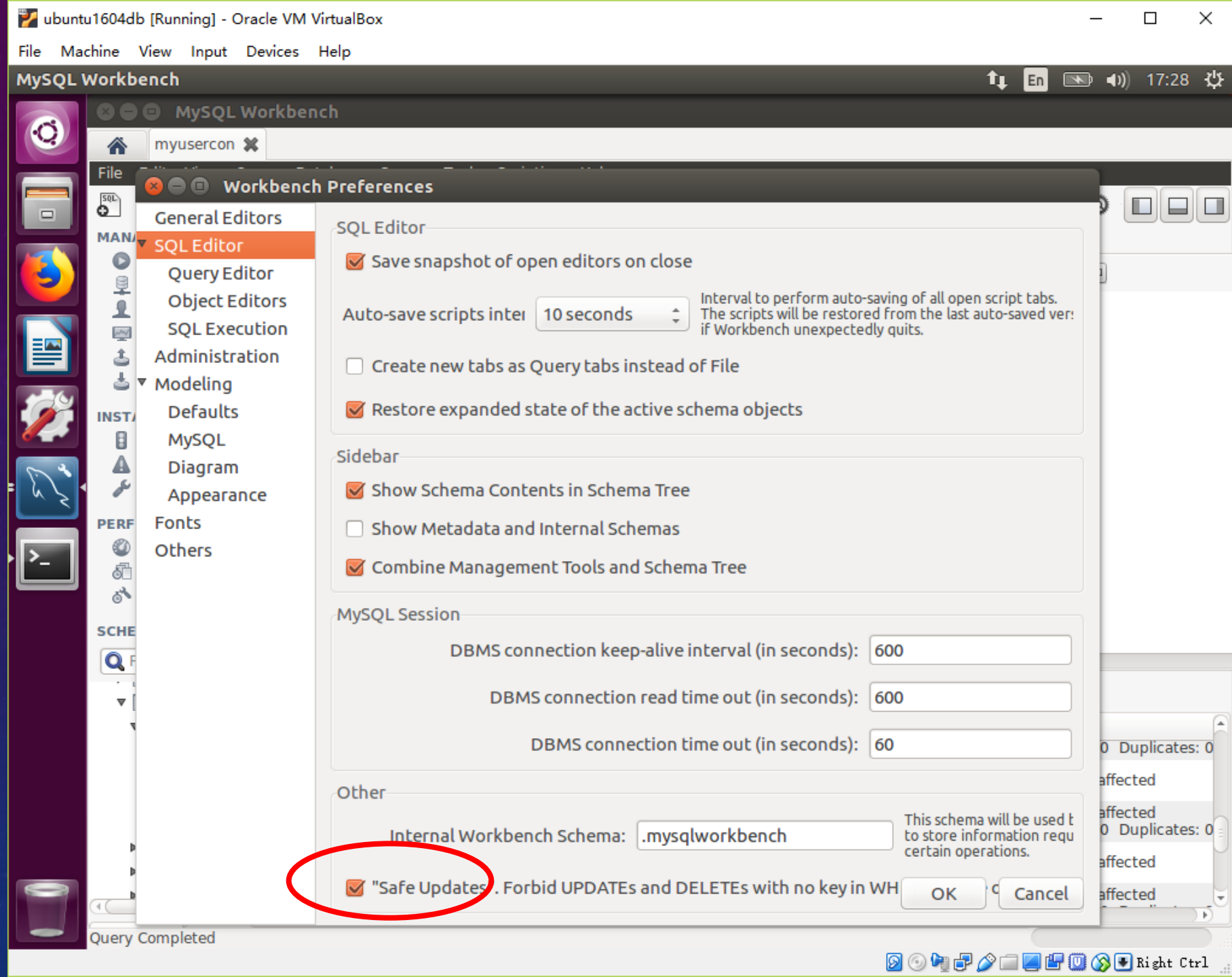
	Time	Action	Message
✓	35	16:43:59	select ID,name,dept_name from instructor LIMIT 0, 1000

0 row

Query Completed

MYSQL: IMPORT DATA OF LABS

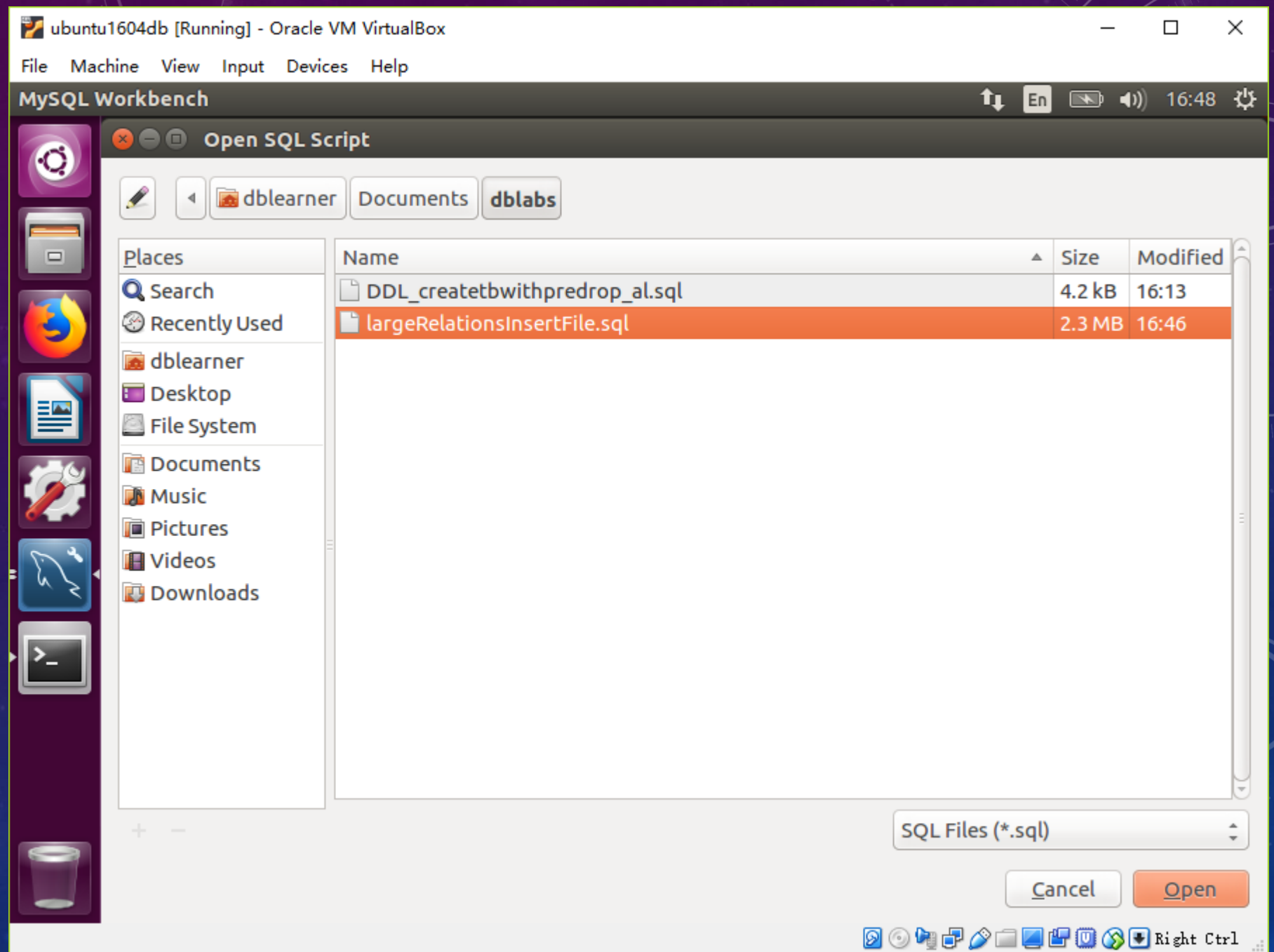
- New DB
 - dbsclab2018
 - Disable Safe Updates
 - Restart workbench
- ©LXD



MYSQL: IMPORT DATA OF LABS

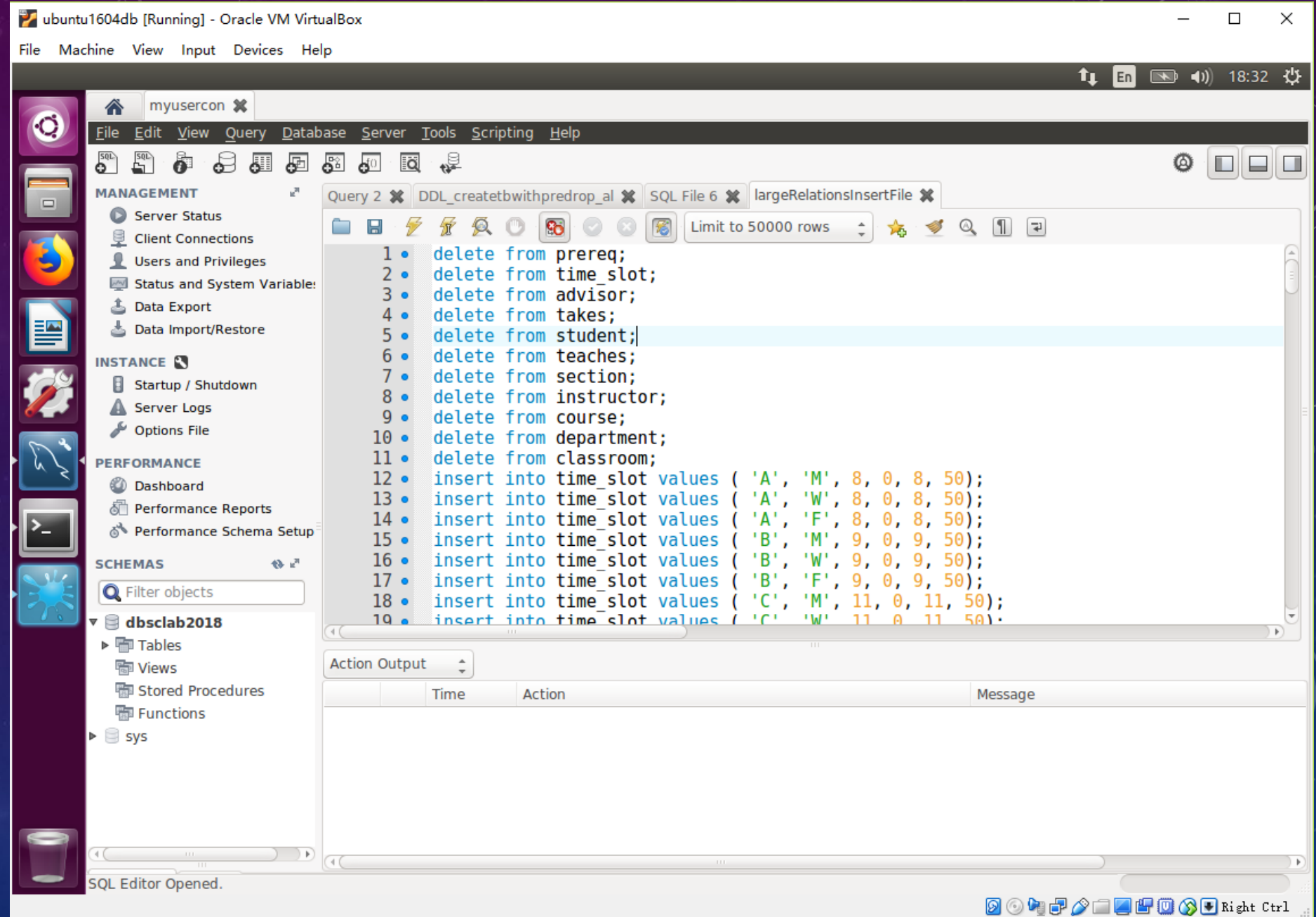
- New DB
 - Dbsclab2018
 - Batch import

©LXD



MYSQL: IMPORT DATA OF LABS

- New DB
 - Dbsclab2018
 - Batch import



MYSQL: IMPORT DATA OF LABS

- New DB
 - Dbsclab2018
 - Batch import

```
$mysql -umyuser -p  
show databases;  
use dbsclab2018;  
source largeRelationsInsertFile.sql;  
commit;  
quit;
```

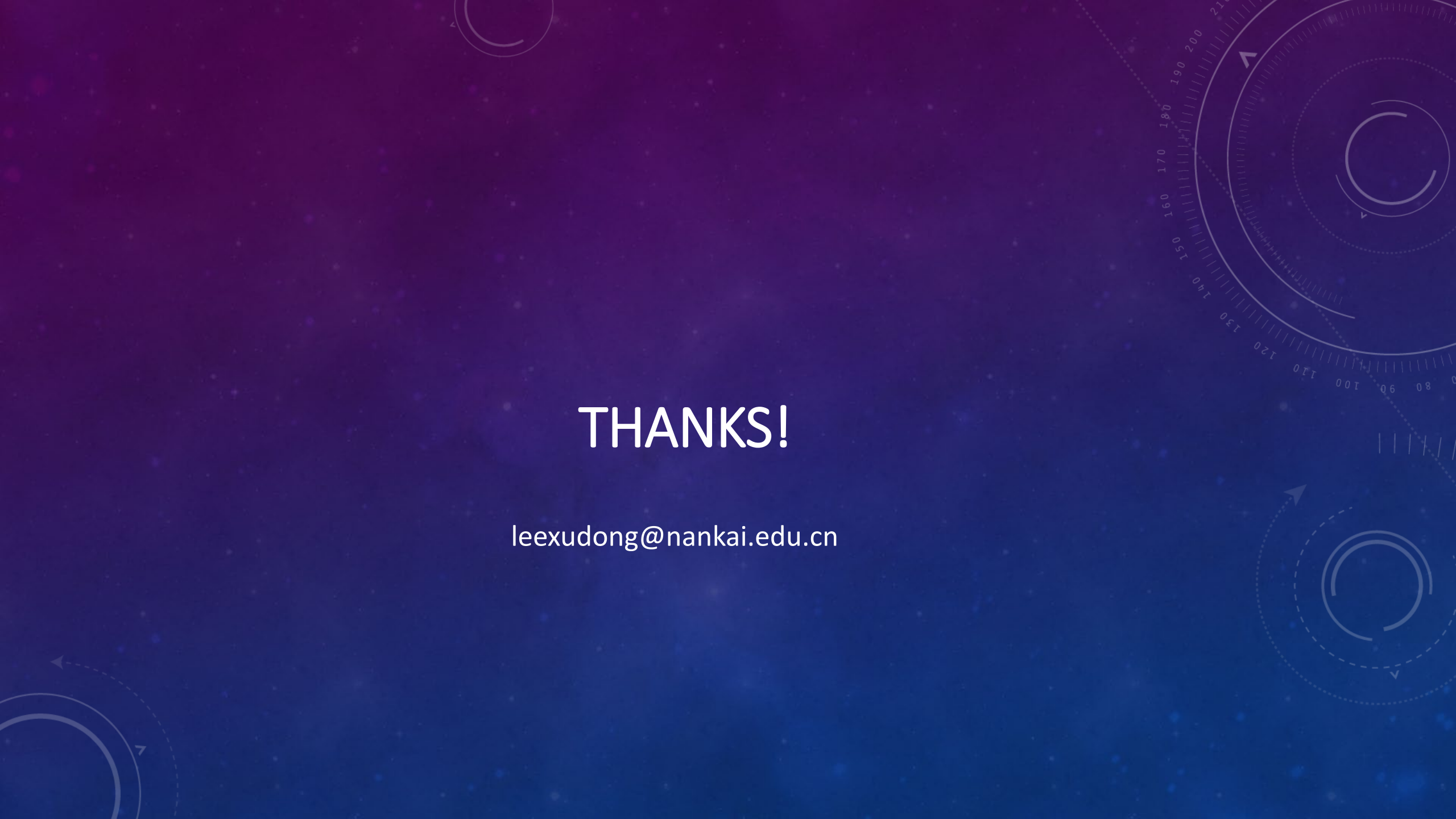
SIMPLE QUERY STATEMENT

SIMPLE QUERY STATEMENT

- `SELECT * FROM department`
- `select dept_name, budget from department`
- `select * from student where dept_name='History'`

QUIZ 4

- 采用两种方法做如下事情：
 - 创建一个员工表，其包括员工工号、姓名、年龄、性别、联系方式、入职时间、工资，其中性别的值为“男”或“女”。



THANKS!

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